

LOGAN-CACHE, UT, USA

WMO: 724796

Lat: 41.787N

Lon: 111.853W

Elev: 4454

StdP: 12.48

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94128

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	1.0	-8.5	4.1	-4.5	-2.9	5.5	2.4	20.9	37.2	17.1	35.0	1.2	230	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.4	94.9	61.6	92.4	61.0	90.0	60.3	65.0	85.3	63.8	84.4	62.4	83.5	6.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	88.5	70.0	56.6	80.7	69.4	54.6	74.9	69.3	32.2	84.7	31.2	83.9	30.2	83.7	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	16.7	13.2	-10.5	98.7	7.4	2.0	-15.8	100.1	-20.2	101.2	-24.3	102.4	-29.7	103.8		
			-11.0	68.4	7.3	3.5	-16.2	70.9	-20.5	73.0	-24.6	75.0	-30.0	77.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	22.3	26.9	38.2	46.3	54.4	63.9	72.8	70.3	60.5	47.6	35.9	24.4	(6)
(7)		DBStd	18.79	11.32	10.89	8.62	6.59	7.31	6.88	4.10	4.50	6.88	7.11	8.35	10.18	(7)
(8)		HDD50	3424	858	648	371	146	40	4	0	0	8	128	426	795	(8)
(9)		HDD65	7061	1323	1068	829	560	332	98	2	11	164	540	874	1260	(9)
(10)		CDD50	2358	0	0	7	37	178	421	708	630	322	53	2	0	(10)
(11)		CDD65	519	0	0	0	0	5	65	245	176	28	0	0	0	(11)
(12)		CDH74	9790	0	0	0	16	289	1588	3956	3011	890	40	0	0	(12)
(13)		CDH80	4590	0	0	0	1	66	652	2110	1472	284	5	0	0	(13)
(14)	Wind	WSAvg	4.4	3.0	3.7	4.8	5.9	5.3	5.1	4.8	4.9	4.2	3.9	3.6	3.3	(14)
(15)	Precipitation	PrecAvg	16.5	1.6	1.3	1.6	1.9	2.0	1.3	0.6	0.8	1.0	1.6	1.3	1.5	(15)
(16)		PrecMax	25.1	4.4	2.7	3.6	5.2	4.9	3.8	2.3	2.5	3.3	4.9	2.6	4.2	(16)
(17)		PrecMin	10.9	0.4	0.1	0.6	0.4	0.4	0.2	0.0	0.0	0.1	0.0	0.3	0.1	(17)
(18)		PrecStd	3.7	1.1	0.7	0.9	1.0	1.3	0.9	0.6	0.6	0.8	1.0	0.6	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	56.6	69.7	76.1	85.9	94.1	98.9	96.1	90.3	79.1	63.8	51.9	(19)
(20)			MCWB	39.4	43.4	49.3	53.0	57.1	61.3	63.4	62.1	60.6	54.0	46.6	42.1	(20)
(21)		2%	DB	43.6	51.0	63.8	71.2	81.4	90.7	96.4	93.3	86.7	72.8	59.1	45.8	(21)
(22)			MCWB	37.7	41.4	46.6	50.0	56.9	59.4	61.9	61.2	58.1	52.1	45.3	39.3	(22)
(23)		5%	DB	39.4	45.6	58.5	66.4	76.8	87.6	93.7	91.0	83.2	68.1	54.3	40.8	(23)
(24)			MCWB	35.2	38.6	44.4	48.3	55.3	59.0	61.5	60.4	56.6	50.5	43.4	36.2	(24)
(25)		10%	DB	36.2	40.9	53.6	62.6	72.3	83.6	91.1	88.6	79.6	63.6	49.3	37.2	(25)
(26)			MCWB	33.4	35.8	42.4	47.0	53.8	58.2	61.1	59.7	55.7	48.5	41.4	34.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	44.4	50.5	54.5	61.0	64.7	67.1	65.7	63.8	56.7	49.5	43.6	(27)
(28)			MDCB	46.8	53.9	65.7	73.3	80.7	86.9	85.5	86.0	79.7	72.1	59.5	50.0	(28)
(29)		2%	WB	38.1	41.7	47.9	51.3	58.2	62.2	65.6	64.3	60.7	54.3	46.3	39.7	(29)
(30)			MDCB	42.2	49.6	61.6	67.0	76.4	84.2	85.7	85.2	79.2	67.4	56.1	45.0	(30)
(31)		5%	WB	35.7	39.0	45.3	49.4	56.3	60.5	64.4	63.1	59.2	52.1	44.1	36.9	(31)
(32)			MDCB	39.4	44.9	57.3	64.1	74.1	81.7	84.7	84.2	77.2	64.6	53.1	40.3	(32)
(33)		10%	WB	33.4	36.3	42.8	47.7	54.3	59.2	63.2	61.8	57.3	49.8	41.8	34.4	(33)
(34)			MDCB	35.8	40.6	51.9	60.7	70.3	79.9	83.8	82.9	75.1	62.5	49.1	37.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.2	17.4	20.8	23.7	26.7	32.3	35.4	34.9	33.0	27.9	21.3	15.4	(35)
(36)			MCDBR	17.3	20.4	30.6	33.7	36.1	39.5	40.0	39.3	39.5	36.6	29.3	18.4	(36)
(37)			MCWBR	12.1	12.6	15.8	16.2	15.5	14.0	12.9	13.0	15.0	18.0	17.0	12.6	(37)
(38)		5% WB	MCDBR	16.3	19.4	29.2	30.4	33.2	35.4	33.9	33.2	32.9	31.9	27.2	17.3	(38)
(39)			MCWBR	11.6	12.1	15.6	15.0	14.8	13.7	11.5	11.8	12.9	15.6	16.4	12.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.268	0.287	0.316	0.330	0.320	0.349	0.349	0.318	0.295	0.267	0.253	(40)	
(41)		taud	2.405	2.422	2.492	2.430	2.440	2.507	2.444	2.427	2.503	2.546	2.558	2.482	(41)	
(42)		Ebn at Noon	289	302	307	303	299	301	291	288	292	288	284	282	(42)	
(43)		Edh at Noon	24	29	30	35	36	33	35	35	30	25	21	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	601	874	1325	1718	2032	2438	2404	2081	1661	1122	706	489	(44)	
(45)		RadStd	65	101	92	107	180	174	108	95	110	75	74	63	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page